Curriculum Vitae/Resume

Xiaoyu Ma

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Research Interests

AI infrastructure; LLM serving and LLM agent systems; GPU programming; cloud computing.

Education

Shanghai Jiao Tong University 2023 – 2027 (expected)
B.Eng. in Computer Science and Engineering
Overall GPA: 3.77/4.00 | Core GPA: 3.80/4.00

Research Experience

Jaseci Lab July 2026 – Present
University of Michigan, Ann Arbor, MI, United States

MORTIS: Compiler-Guided KV-Cache Management for LLM Agent Programs
First author | [S1]

- Designed and implemented a compiler–serving co-design system that identifies and exploits differences in sharing scope, lifetime, and evolution across prompt components.
- Used program-derived prompt structure to guide prompt reordering for greater cache reuse, and to retain or prefetch KV cache for upcoming LLM calls.
- Achieved up to 1.99× job completion time speedup and 1.97× throughput improvement over SGLang.

Prompt Less, Smile More: Semantic Engineering in Lieu of Prompt Engineering
Co-author | [S2]

- Conducted system benchmarking and evaluation, and contributed to the paper's methodology analysis.

Agent-Native by Design: A Programming Language for Agentic Applications
Co-author | [P1]

- Built experimental agentic workloads across Jac and baseline frameworks to benchmark end-to-end system performance.

Emerging Parallel Computing Center (EPCC) June 2025 – June 2026
Shanghai Jiao Tong University, Shanghai, China

SLO-Aware Proactive History Compression for Long-History Generative Recommendation Serving
First author | [C1]

- Designed and implemented an SLO-aware GPU spatial multiplexing mechanism built on CUDA green contexts, enabling proactive, asynchronous history compression alongside live serving of HSTU generative recommendation models.
- Developed an SLO-aware scheduler that dynamically utilizes idle GPU capacity under fluctuating workloads, increasing system goodput by 29.6%.

ZAP: Adaptive Templating for Fast Startup of LLM Functions on GPUs
Contributor

- Implemented GPU P2P and RDMA-based model transfer; optimized weight loading with multithreading and fixed correctness bugs.

Publications & Manuscripts

Conference Papers

[C1] **Xiaoyu Ma**, Weihao Cui. "SLO-Aware Proactive History Compression for Long-History Generative Recommendation Serving." In *Proceedings of the ACM Symposium on Cloud Computing (SoCC)*, 2026.

Under Review

[S1] **Xiaoyu Ma**, Jayanaka L. Dantanarayana, Savini Kashmira, Baichuan Li, Lingjia Tang, Jason Mars. "MORTIS: Compiler-Guided KV-Cache Management for LLM Agent Programs." Under review at *MLSys* 2027.

[S2] Jayanaka L. Dantanarayana, Savini Kashmira, **Xiaoyu Ma**, Krisztian Flautner, Lingjia Tang, Jason Mars. "Prompt Less, Smile More: Semantic Engineering in Lieu of Prompt Engineering." Under review at *OOPSLA* 2027.

In Preparation

[P1] Jayanaka L. Dantanarayana, Savini Kashmira, **Xiaoyu Ma**, M. R. Malitha Prabhashana, Lingjia Tang, Jason Mars. "Agent-Native by Design: A Programming Language for Agentic Applications." In preparation for *PLDI* 2027.

Technical Skills

Languages	C/C++, Python, CUDA, Jac
Systems & AI Infrastructure	GPU programming, RDMA, LLM/agent serving, runtime scheduling

Honors & Awards

SJTU Academic Excellence Scholarship 2025